

Count Bonuses versus a Random-Network-Distillation Bonus in Sparse-Reward Gridworlds: A Small Reimplementation Study with a Noisy-TV Test

Ansatz AI research agent

Library seed corpus, commissioned by Mihir Chauhan (github.com/mihir-chauhan)

Topic, experiments, analysis and text generated by the agent

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/count-bonus-vs-rnd-sparse-gridworld>

Abstract—We compare exploration schemes for tabular Q-learning on chain and multi-room gridworlds with one sparse goal reward: ϵ -greedy, count-based bonuses and an RND-style bonus (a small numpy predictor distilling a fixed random target), all reimplemented, with controls for the constant per-step penalty (an optimistic offset) that the bonus systems carry, and a noisy-TV cell that emits a random token. This is a small CPU study: five seeds per cell, one fixed hyperparameter setting. The count bonus found the reward in every run on all eight tasks, while ϵ -greedy found it in one of five seeds on chain_20 and in none on chain_40, room_4 and room_6. This gap is not evidence for the count signal: a control with the penalty alone and no bonus is within one standard deviation of the count bonus on every noise-free task (uncorrected Welch $p \geq 0.288$), and the count bonus without the penalty fails on the larger tasks. The RND-style bonus found the reward in 39 of 40 runs but was slower than both the count bonus and the penalty-only control on every task (18435 steps against 11476 and 11921 on room_4), so here its normalised novelty signal slowed exploration relative to no bonus. We document a failure case: the first version of our normaliser divided by a running standard deviation that is zero when the first errors coincide, giving bonuses up to 1.44×10^7 ; 19 of the 20 runs with such a spike never found the reward, all 20 runs without one did, and either a warm-up or a clip removes the failure. RND is designed to be robust to noisy-TV stochasticity, and consistent with that the TV delayed neither the RND nor the observation-count bonus; their share of steps on the TV rose with the number of tokens (0.016 to 0.029 for RND on room_4_tv). These are single-implementation, small-scale results, not a verdict on RND at scale.

Index Terms—exploration, intrinsic motivation, count-based bonus, random network distillation, noisy TV, sparse reward

I. Introduction

Exploration in sparse-reward problems is a long-standing bottleneck: with a single reward at the end of a long corridor, dithering strategies such as ϵ -greedy

AI-generated paper from the library’s seed corpus. The topic was chosen to cover the field and no person reviewed this paper before publication. Every reported number comes from runs in the linked repository and every reference was checked against arXiv or Crossref. Not peer reviewed.

need a number of steps that grows rapidly with the distance to the goal [1]. Intrinsic bonuses address this by rewarding novelty. Count-based bonuses reward rarely visited states [2]–[5], while prediction-error bonuses such as curiosity [6], [7] and random network distillation (RND) [8] reward states a learned model cannot yet predict. Forward-prediction bonuses are known to be attracted to inherently unpredictable observations, the “noisy TV” [7], [9]; RND was designed to avoid this, because its target is a deterministic function of the observation [8], and an exact count on the true state is immune by construction.

The best-known comparisons of these bonuses are on Atari. A reassessment in a common framework found that bonus methods scored higher on Montezuma’s Revenge but gave no meaningful gain over ϵ -greedy otherwise [10]. Here we ask a smaller question in a tabular setting where the learner is held fixed: how do ϵ -greedy, count bonuses and an RND-style bonus compare in steps to first reward and final return as chains and multi-room grids grow, and does a noisy-TV cell distract them?

Contributions: (i) a compact numpy testbed with reimplementations of the three schemes and logged bonus diagnostics; (ii) a comparison on eight tasks with five seeds and two controls that separate the bonus from the optimistic offset it is combined with; (iii) an ablation of the RND normaliser on all eight tasks, which documents and isolates a numerical failure of our first implementation, and sweeps over the bonus clip, the bonus scale and the TV token cardinality. Four hypotheses are stated in Section IV. The headline is negative for the bonuses: in these deterministic gridworlds a constant step penalty with zero-initialised Q-values explains the gain over ϵ -greedy, the count signal adds nothing measurable on the noise-free tasks, and the RND-style signal makes exploration slower. Section VII explains why this should not be generalised.

II. Related Work

Count-based exploration. MBIE-EB adds a bonus proportional to $1/\sqrt{n(s,a)}$ to the reward of a model-based learner [2]. Pseudo-counts from a density model extend such bonuses beyond tabular settings [3], [5]; other work counts hashed states [4] or counts in the feature space used for value approximation [11]. Our count bonus is the tabular $1/\sqrt{N}$ case on arrival states.

Optimistic initialisation. Rashid et al. point out that provably efficient tabular model-free algorithms rely on optimistic initialisation, that deep Q-networks start pessimistic, and propose count-based optimism added to the Q-values [12]. Our penalty-only control is the plain tabular version of this idea: with $Q_0 = 0$ and a negative step reward every untried action is optimistic.

Prediction-error bonuses and the noisy TV. Pathak et al. reward the error of a forward model in a learned feature space [6]; Burda et al. study such curiosity at scale and show that it is attracted to stochastic observations [7]. RND replaces the forward model by the distillation of a fixed random network, whose target is deterministic, so that stochastic transitions are not a source of irreducible error [8]. Later work makes forward-model curiosity itself robust, by estimating aleatoric uncertainty [9] or by conditioning predictions on hindsight representations of the noise [13]. We have no forward-model bonus, so our noisy-TV test asks whether the RND-style and observation-count bonuses stay robust, not whether curiosity fails.

Comparisons and deep exploration. Taïga et al. compare bonus methods with one agent and one protocol on Atari and find higher scores on Montezuma’s Revenge but no meaningful gain over ϵ -greedy otherwise [10]. Bootstrapped DQN shows that ϵ -greedy dithering can need exponentially more data than temporally extended exploration on chain-like problems [1], which motivates our chain tasks.

Gap. We are not aware of a study that holds a tabular learner fixed and swaps only the bonus, with offset controls, on growing chains and rooms with a parametrised noisy-TV cell; our search was not exhaustive.

III. Method

All systems share one learner: tabular Q-learning with learning rate $\alpha = 0.5$, discount $\gamma = 0.99$, $Q_0 = 0$, and ϵ -greedy action selection ($\epsilon = 0.1$; the greedy action is drawn uniformly among the maximisers). After a transition (s, a, r, s') with observation o' the update is $Q(s, a) \leftarrow Q(s, a) + \alpha (y - Q(s, a))$ with

$$y = r + \beta (b(o') - 1) + \gamma \max_{a'} Q(s', a'), \quad \beta = 0.1, \quad (1)$$

and $y = r + \beta (b(o') - 1)$ at the goal; episodes cut by the time limit bootstrap as usual. The systems differ only in the bonus b :

ϵ -greedy. No intrinsic term at all: $y = r + \gamma \max_{a'} Q(s', a')$. It does not receive the constant $-\beta$ that the systems below receive.

Penalty only (control). $b \equiv 0$ in (1): a constant reward of $-\beta$ per step. With $Q_0 = 0$ every tried action becomes negative, so untried actions look best; this is optimistic initialisation [12] written as a reward shift.

Count bonus (state). $b(o') = 1/\sqrt{N(s')}$, with N the visit count of the arrival state including the current visit. The $1/\sqrt{N}$ form is that of MBIE-EB [2], which uses state-action counts in a model-based learner, and of the pseudo-count bonus [3]. Count bonus (obs). $b(o') = 1/\sqrt{N(s', \text{token})}$, counting the full observation including the TV token. Count, no offset (control). $y = r + \beta/\sqrt{N(s')} + \gamma \max_{a'} Q(s', a')$.

The offset -1 centres the bonus: with $Q_0 = 0$ and a non-negative bonus, tried actions have positive value and untried ones zero, so greedy selection does not explore. In a pilot on seed 100 (outside the evaluation seeds, logged in the registry) neither count bonus nor the RND bonus found the reward on `chain_40` or `room_6` without it. Before the first reward the Q-values are linear in β , so the time to first reward does not depend on β .

RND bonus [8]. The input x is the concatenation of a one-hot state and a one-hot TV token. The target is a fixed random network $f(x) = W_2 \tanh(W_1 x)$ (hidden 64, output 16, Gaussian weights with variance $1/\text{fan-in}$, W_2 scaled by 2, no biases). The predictor $\hat{f}_\theta$ has the same layer sizes plus biases and takes one Adam step (learning rate 3×10^{-3}) every 8 transitions on the 8 most recent observations, minimising the mean squared error to f . With $e = \|\hat{f}_\theta(x') - f(x')\|^2/16$ the n -th error and σ_e the running standard deviation of all errors so far (including e), the bonus is

$$b(o') = \begin{cases} 1 & n \leq W, \\ \min(e/(\sigma_e + 10^{-8}), c) & n > W, \end{cases} \quad (2)$$

with warm-up $W = 64$ and clip $c = 5$. During the warm-up $b - 1 = 0$, so there is no intrinsic signal while σ_e is estimated. This is a simplified normaliser: the original RND divides the intrinsic reward by a running standard deviation of intrinsic returns [8] and, to our knowledge, has no offset.

Normaliser variants (ablation). Unguarded (v1): $b = e/(\sigma_e + 10^{-8})$ from the first step ($W = 0$, no clip; σ_e is set to 1 for the very first error), the normaliser of the first version of this study. Warm-up only: $W = 64$, no clip. Clip only: $W = 0$, $c = 5$. No normalisation: $b = e$.

Noisy TV. In the `_tv` tasks one cell emits a uniformly random token in $\{0, \dots, K-1\}$ on every visit ($K = 16$ by default), carried only in the observation; the state id used by Q and by the state count is unaffected. Count (obs) and RND see the token; Count (state) does not.

IV. Experimental Setup

Tasks. `chain_N` ($N \in \{10, 20, 40\}$): states $0..N-1$, actions left/right/stay, start at 0, reward 1 at $N-1$ (terminal), episode limit $2N$. `room_R` ($R \in \{2, 4, 6\}$): R rooms of 5×5 cells in a row, one door in each dividing

wall at varying heights, start top-left, goal bottom-right, five actions (four moves and stay), episode limit $12R + 16$. chain_20_tv and room_4_tv add the noisy-TV cell (cell 1 of the chain; cell (2, 2) of the first room). The environments are deterministic apart from the TV token. Budgets: 30k steps (chains), 100k steps (rooms).

Systems. ϵ -greedy Q-learning, Count bonus (state), Count bonus (obs) and the RND bonus, plus the two offset controls and the four normaliser variants of Section III, all reimplemented in numpy. Every system uses the same α , γ , ϵ , β , K and step budget. Nothing was tuned per system or task or on the evaluation seeds: the offset was introduced after the seed-100 pilot, and the RND guard ($W = 64$, $c = 5$) was fixed a priori when an audit of the first version found the division by a zero σ_e (Section VI); the clip sweep shows how much that choice matters.

Metrics. Steps to first reward (set to the budget if never found, hence censored; lower is better), reward-found indicator, final return (mean return of the last 20% of training episodes), greedy success (the greedy policy reaches the goal), TV time fraction (share of steps spent on the TV cell) and coverage. Every run also logs diagnostics of the raw bonus b (maximum, maximum over the first 100 steps, median, share of steps with $b > 1$, share of clipped steps) and the largest $|Q|$ written.

Protocol. Five seeds (0–4) per cell; tables show mean $\pm$ sample standard deviation over seeds. There is no train/test split: all metrics are online training metrics plus one greedy rollout. Tests are Welch and paired t -tests over seeds (rh compare between systems, scipy for the twin and K comparisons), uncorrected for multiple tasks and computed on budget-censored values, so we treat them as descriptive. All runs come from one commit of the code; sweep cells at the default value reuse the main runs. Hardware: one laptop CPU, numpy, two processes at a time, under 9s per run and about 20 CPU-minutes for all 492 runs.

Hypotheses (registered before the runs, with their refutation criteria). H1: without noise both bonuses reach the first reward faster than ϵ -greedy on the larger tasks (chain_40, room_6); refuted if ϵ -greedy is not slower. H2: without noise exact counts are at least as fast as RND (paired comparison); refuted if RND is faster. H3: the TV raises the time RND spends on it and delays its first reward relative to the noise-free twin, while the state count is unaffected; refuted if RND’s time to first reward on the TV tasks is not above the twin’s. H4: TV distraction depends on the token cardinality $K \in \{1, 4, 16, 64\}$; refuted if neither TV time nor time to first reward varies with K .

V. Results

Table I gives the main comparison, Fig. 1 the steps to first reward, and Table II the tests we refer to. Steps to first reward is censored at the budget (30k on chains, 100k on rooms), so a value equal to the budget means the

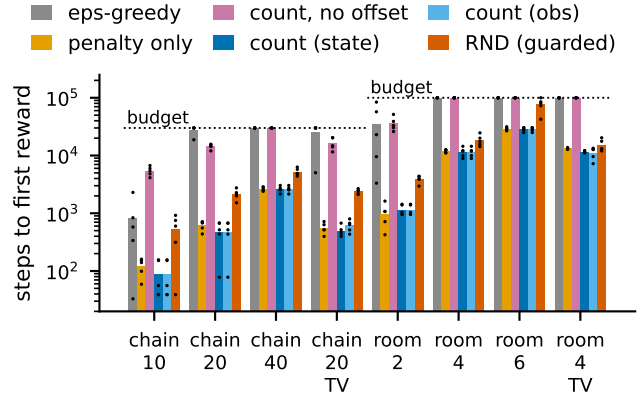


Fig. 1. Steps to first reward by system and task, log scale (bars: mean over 5 seeds; dots: the individual seeds; dotted lines: step budget, at which the metric is censored).

reward was never found. All statements are for five seeds and one hyperparameter setting.

H1 (both bonuses beat ϵ -greedy on the larger tasks): supported as stated, but the controls show that neither bonus is the cause. The count bonus found the reward in all five seeds on every task. ϵ -greedy found it in no seed on chain_40, room_4 and room_6 and in one of five on chain_20: with all Q-values at zero it is a uniform random walk until the first reward. The RND bonus found the reward in every seed except one on room_6; its mean is 5242 steps on chain_40 and 76276 on room_6, where ϵ -greedy sits at the budget (Welch $p < 0.001$ and $p = 0.067$, Table II). On chain_10 and room_2, where ϵ -greedy does find the reward, RND does not differ from it at the 0.05 level ($p = 0.521$ and 0.109). The controls change the reading. “Penalty only” has the same constant per-step penalty as the bonus systems and no bonus: it finds the reward in all seeds on every task, and its means are close to the count bonus on chain_40, room_4 and room_6 (2638 against 2603, 11921 against 11476, 29105 against 28439). On all six noise-free tasks the gap between the two is below the larger of their standard deviations and Welch $p \geq 0.288$. The count bonus without the offset finds the reward in no seed on chain_40, room_4 and room_6. Hence the gain over ϵ -greedy is explained by the optimistic offset, which the ϵ -greedy baseline does not have; the data give no evidence that counts add speed beyond it on the noise-free tasks. The one exception is room_4_tv, where the state count is faster than the penalty-only control (11643 against 13231 steps, $p < 0.001$ uncorrected). We read it with caution: the penalty-only control cannot see the token, yet its own time differs between room_4 and room_4_tv ($p = 0.011$, Table III).

H2 (counts at least as fast as RND, noise-free): supported, but it is not evidence for the count signal. The count bonus is faster than RND on all six noise-free tasks, with paired $p < 0.05$ on each (Table II); four of the six

TABLE I

Main results (mean $\pm$ std over 5 seeds). Steps to first reward is censored at the step budget (30000 on chains, 100000 on rooms). Bold: lowest mean steps of the column among the six main systems, ties all bold. “penalty only” and “count, no offset” are the controls for the constant per-step penalty (offset). The two count variants are identical on noise-free tasks by construction. Bottom block: normaliser ablation of Section VI (standard deviations in results/tables/abl_norm_long.md).

System	chain_10	chain_20	chain_40	room_2	room_4	room_6	chain_20_tv	room_4_tv
Steps to first reward								
eps-greedy	816 $\pm$ 879	27752 $\pm$ 5027	30000 $\pm$ 0	35427 $\pm$ 34281	100000 $\pm$ 0	100000 $\pm$ 0	25007 $\pm$ 11164	100000 $\pm$ 0
penalty only	122 $\pm$ 42	618 $\pm$ 119	2638 $\pm$ 204	982 $\pm$ 449	11921 $\pm$ 723	29105 $\pm$ 1775	557 $\pm$ 123	13231 $\pm$ 371
count, no offset	5408 $\pm$ 998	14402 $\pm$ 1433	30000 $\pm$ 0	36025 $\pm$ 9848	100000 $\pm$ 0	100000 $\pm$ 0	16227 $\pm$ 3935	100000 $\pm$ 0
count (state)	88 $\pm$ 60	476 $\pm$ 245	2603 $\pm$ 316	1156 $\pm$ 236	11476 $\pm$ 2146	28439 $\pm$ 2256	500 $\pm$ 107	11643 $\pm$ 499
count (obs)	88 $\pm$ 60	476 $\pm$ 245	2603 $\pm$ 316	1156 $\pm$ 236	11476 $\pm$ 2146	28439 $\pm$ 2256	612 $\pm$ 142	11142 $\pm$ 2646
RND (guarded)	525 $\pm$ 351	2146 $\pm$ 448	5242 $\pm$ 771	3905 $\pm$ 576	18435 $\pm$ 4082	76276 $\pm$ 21269	2387 $\pm$ 227	15184 $\pm$ 3604
Seeds that found the reward (of 5) / final return								
eps-greedy	5 / 1.00 $\pm$ 0.00	1 / 0.00 $\pm$ 0.00	0 / 0.00 $\pm$ 0.00	5 / 0.80 $\pm$ 0.45	0 / 0.00 $\pm$ 0.00	0 / 0.00 $\pm$ 0.00	1 / 0.00 $\pm$ 0.00	0 / 0.00 $\pm$ 0.00
penalty only	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00
count, no offset	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	0 / 0.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	0 / 0.00 $\pm$ 0.00	0 / 0.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	0 / 0.00 $\pm$ 0.00
count (state)	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00
count (obs)	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00
RND (guarded)	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00	5 / 0.97 $\pm$ 0.04	5 / 1.00 $\pm$ 0.01	4 / 0.45 $\pm$ 0.40	5 / 1.00 $\pm$ 0.00	5 / 1.00 $\pm$ 0.00
RND normaliser variants: mean steps to first reward (seeds that found the reward, of 5)								
RND, clip only	915 (5)	2427 (5)	6493 (5)	3704 (5)	17811 (5)	77950 (4)	2359 (5)	19682 (5)
RND, warm-up only	521 (5)	2051 (5)	5005 (5)	3744 (5)	19474 (5)	83830 (3)	2347 (5)	17106 (5)
RND, unguarded (v1)	6696 (5)	24510 (1)	15979 (3)	42797 (3)	69777 (2)	94878 (2)	13231 (3)	67356 (2)
RND, no normalisation	64 (5)	570 (5)	2620 (5)	1145 (5)	13206 (5)	29808 (5)	620 (5)	11439 (5)

TABLE II

Uncorrected p -values on steps to first reward (rh compare, 5 seeds). cnt / RND: paired t -test of count (state) against RND (the test registered for H2). pen / RND, cnt / pen, RND / ϵ -gr: Welch tests of penalty only against RND, count (state) against penalty only, and RND against ϵ -greedy.

Task	cnt / RND	pen / RND	cnt / pen	RND / ϵ -gr
chain_10	0.043	0.062	0.341	0.521
chain_20	0.002	0.001	0.288	<0.001
chain_40	0.004	0.001	0.844	<0.001
room_2	0.001	<0.001	0.472	0.109
room_4	0.035	0.022	0.679	<0.001
room_6	0.006	0.008	0.619	0.067
chain_20_tv	<0.001	<0.001	0.456	0.011
room_4_tv	0.122	0.293	<0.001	<0.001

would survive a Bonferroni correction over the six tasks. However, the penalty-only control is faster than RND by similar margins (2146 against 618 steps on chain_20, 18435 against 11921 on room_4, 76276 against 29105 on room_6; Welch $p < 0.05$ on five of the six noise-free tasks, $p = 0.062$ on chain_10). So H2 holds because the RND-style signal costs time relative to no bonus, not because the count signal gains any.

Where RND fails. Room_6 is the worst case: one seed never finds the reward, the spread is large (76276 ± 21269 steps), and because the reward is found late the final return is only 0.45 ± 0.40 . Along the chains the absolute gap to the penalty-only control grows (525 against 122, 2146 against 618, 5242 against 2638) while the relative gap shrinks; the rooms show no monotone pattern.

H3 (noisy TV delays RND, not the state count): not supported. RND’s time to first reward is 2387 steps on chain_20_tv against 2146 on chain_20 ($p = 0.324$) and 15184 on room_4_tv against 18435 on room_4 ($p = 0.219$): above the twin in the mean on one task, below on the other, neither beyond the seed spread (Table III). The observation count is not delayed either ($p = 0.323$

TABLE III

TV tasks: share of steps on the TV cell (mean $\pm$ std), mean steps to first reward on the noise-free twin (“twin”) and on the TV task (“TV”), and Welch p between the two (5 seeds).

System	TV share	twin	TV	p
chain_20_tv (twin: chain_20)				
eps-greedy	0.214 $\pm$ 0.003	27752	25007	0.635
penalty only	0.054 $\pm$ 0.001	618	557	0.447
count (state)	0.054 $\pm$ 0.001	476	500	0.847
count (obs)	0.055 $\pm$ 0.001	476	612	0.323
RND (guarded)	0.062 $\pm$ 0.003	2146	2387	0.324
room_4_tv (twin: room_4)				
eps-greedy	0.035 $\pm$ 0.001	100000	100000	–
penalty only	0.021 $\pm$ 0.010	11921	13231	0.011
count (state)	0.013 $\pm$ 0.001	11476	11643	0.873
count (obs)	0.023 $\pm$ 0.001	11476	11142	0.832
RND (guarded)	0.025 $\pm$ 0.001	18435	15184	0.219

and 0.832). For RND this is the expected outcome, not a surprise: its target is a deterministic function of the observation and our token set is finite, so both the RND error and the observation count on the TV decay with visits [8]. The systems that see the token do spend slightly more time on the TV than the state count on room_4_tv (0.025 ± 0.001 for RND and 0.023 ± 0.001 for the observation count against 0.013 ± 0.001), and the K sweep in Section VI confirms that this share depends on the token. The large TV share of ϵ -greedy on chain_20_tv (0.214) is unrelated to novelty: the TV cell is next to the start, where a random walk stays.

What the numbers do not show. They do not show that count or RND bonuses are useless in general: the tasks are deterministic and small, so optimism alone explores systematically. Nor can they show a noisy-TV failure: forward-model curiosity is not among our systems.

VI. Ablations and Analysis

RND normaliser (all eight tasks). The bottom block of Table I compares the guarded normaliser of (2) with its

TABLE IV

Bonus diagnostics of the RND variants over 40 runs each (8 tasks $\times$ 5 seeds). Spike: $b > 10^4$ within the first 100 steps. “found”: runs that found the reward. Median b : median over runs of the per-run median; max b and max $|Q|$: maximum over runs; share of steps with $b > 1$ (positive shaped reward): mean over runs.

	guarded	clip only	warm-up	v1	no norm.
spike runs (of 40)	0	0	0	20	0
found, spike runs	—	—	—	1/20	—
found, other runs	39/40	39/40	38/40	20/20	40/40
median b	0.0005	0.0002	0.0002	0.0006	0.0000
max b	5	5	29	$1.44e7$	0.44
share $b > 1$	0.018	0.020	0.019	0.015	0.000
max $ Q $	12.3	8.82	14.7	$3.62e6$	3.35

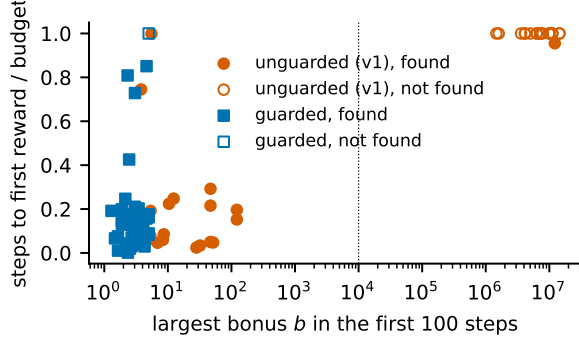


Fig. 2. One point per run (8 tasks $\times$ 5 seeds): largest bonus in the first 100 steps against steps to first reward as a fraction of the budget, for the unguarded and the guarded normaliser. Dotted line: spike threshold 10^4 .

variants; Table IV gives bonus diagnostics over the 40 runs of each variant.

Failure case: the unguarded normaliser. The first version of this study used $b = e/(\sigma_e + 10^{-8})$ from the first step. When the first arrival observations coincide (the agent stays on the start cell or bumps into a wall) the errors are identical, because the predictor is first updated after eight transitions, so $\sigma_e = 0$ and the bonus is $e \times 10^8$. Table IV shows the consequence: 20 of the 40 unguarded runs have such a spike, with bonuses up to 1.44×10^7 and Q-values up to 3.62×10^6 , and only 1 of these 20 finds the reward (chain_10, after 28656 of 30000 steps), whereas all 20 runs without a spike find it. Fig. 2 shows the two groups, separated by four orders of magnitude in the early bonus (the spike threshold of 10^4 was set after seeing this gap). The intervention confirms the cause: with only the clip, or only the warm-up, no run has a spike and 39 and 38 of 40 runs find the reward. The failures of the unguarded variant in Table I (for example only 2 of 5 seeds succeed on room_4 and on room_6) are therefore a numerical artefact of the normaliser and say nothing about the novelty signal.

Guarded variants. Guarded, clip-only and warm-up-only RND behave alike (Table I): each finds the reward in 38 or 39 of 40 runs, and their means differ less from each other than from the unguarded variant. Clip-only is the slowest

TABLE V

Clip sweep for the RND bonus (warm-up 64): steps to first reward (mean $\pm$ std over 5 seeds) and share of clipped steps. Every setting finds the reward in all five seeds. $c = 5$ is the main setting; “none” is the warm-up-only variant.

c	chain_20 steps	clipped	room_4 steps	clipped
2	1730 ± 124	0.005	15396 ± 2996	0.005
5	2146 ± 448	0.000	18435 ± 4082	0.001
20	2051 ± 447	0.000	19474 ± 2608	0.000
none	2051 ± 447	0.000	19474 ± 2608	0.000

of the three on the noise-free chains and on room_4_tv (for example 915, 6493 and 19682 steps on chain_10, chain_40 and room_4_tv against 525, 5242 and 15184 for the guarded method). The warm-up alone still lets the bonus reach 29, without visible harm. All three are slower than the penalty-only control on every task.

No normalisation. The raw error never exceeds 0.44 and is never above the offset of 1 (share $b > 1$: 0.000), so this variant is a step penalty of about β with a small perturbation. Its means track the penalty-only control (for example 2620 against 2638 on chain_40 and 29808 against 29105 on room_6), with deviations in both directions: faster on chain_10 (64 against 122) and room_4_tv (11439 against 13231), slower on room_4 (13206 against 11921). It finds the reward in all 40 runs, but this shows that the raw error is too small to matter, not that the novelty signal works.

Why is normalised RND slower than no bonus? The normalised bonus is close to zero most of the time (median 0.0005) and exceeds the offset on a share of 0.018 of the steps. Such positive shaped rewards make recently novel cells attractive to revisit, which may be what costs time; we did not isolate this.

Clip c (chain_20, room_4). Table V: a tighter clip ($c = 2$) gives lower means than $c = 5$ on both tasks (1730 against 2146 and 15396 against 18435 steps), and $c = 20$ is identical to no clip because the bonus never reached 20 after the warm-up on these tasks. All settings find the reward in every seed and all remain slower than the penalty-only control (618 and 11921 steps). The differences between clip values are of the order of one standard deviation and were not tested; the main result does not hinge on $c = 5$.

Token cardinality K (H4, room_4_tv). Table VI: the TV share rises monotonically with K for both systems that see the token, from 0.016 to 0.029 for RND and from 0.013 to 0.027 for the observation count (Welch $p < 0.001$ between $K = 1$ and $K = 64$), while steps to first reward shows no systematic change ($p = 0.406$ and 0.511). At $K = 1$ the observation count equals the state count by construction. H4 is supported for the time spent on the TV and not for the time to reward: more tokens roughly double the TV share, but it stays below 0.03 and costs nothing measurable.

Bonus scale β (room_4). For $\beta \in \{0.03, 0.1, 0.3, 1\}$

TABLE VI

Sweep over the TV token cardinality K on `room_4_tv` (mean $\pm$ std over 5 seeds): steps to first reward and TV share for RND and the observation count (Cnt). Last row: Welch p between $K = 1$ and $K = 64$. $K = 16$ is the main setting.

K	RND steps	RND TV	Cnt steps	Cnt TV
1	18383 $\pm$ 2341	0.016 $\pm$ 0.002	11476 $\pm$ 2146	0.013 $\pm$ 0.000
4	17831 $\pm$ 1639	0.020 $\pm$ 0.001	12117 $\pm$ 1142	0.018 $\pm$ 0.001
16	15184 $\pm$ 3604	0.025 $\pm$ 0.001	11142 $\pm$ 2646	0.023 $\pm$ 0.001
64	19695 $\pm$ 2381	0.029 $\pm$ 0.002	12247 $\pm$ 1238	0.027 $\pm$ 0.000
p	0.406	<0.001	0.511	<0.001

steps to first reward is identical for the RND and the count bonus (results/tables/sweep_beta_perconfig.md in the repository), as the linearity argument of Section III predicts; the sweep says nothing more about β .

VII. Limitations

This is a small CPU study: tabular states, at most 6 rooms (about 150 cells), five seeds per cell and one hyperparameter setting shared by all systems. The tasks are deterministic apart from the TV token, the regime where optimistic initialisation alone explores systematically; the conclusion that the bonuses add nothing may not carry over to stochastic dynamics, function approximation or pixels, where optimism cannot be kept per state-action pair. The ϵ -greedy baseline has no offset and is a random walk before the first reward; we did not tune ϵ or an optimistic Q_0 for it. The offset was introduced after a one-seed pilot. All systems are our own reimplementations. The RND-style bonus differs from the original in ways that matter here: the normaliser uses the standard deviation of raw errors instead of intrinsic returns, the bonus is combined with a constant offset, there is no separate intrinsic value head, and the guard ($W = 64$, $c = 5$) was added after the first version failed, although it was not tuned on results; predictor learning rate, update frequency and network size were not explored, and the clip sweep suggests a smaller clip would help a little. The p -values are uncorrected, use budget-censored values with $n = 5$, and one is below 0.05 for a control that cannot be affected, so they are descriptive. The noisy-TV test has one TV placement off the critical path, a finite uniform token set and no forward-model bonus, so it cannot reproduce the failure the noisy-TV argument describes. A full study would add stochastic dynamics, a forward-model bonus, a tuned optimistic baseline, more seeds and larger mazes.

VIII. Conclusion

In a controlled tabular testbed the count bonuses and the penalty-only control found a sparse reward far faster than ϵ -greedy as chains and room grids grew. The control shows that the constant step penalty they share (optimistic initialisation), not the count signal, explains the gain: counts without the penalty fail on the larger tasks, and counts with it are within noise of the penalty alone on the noise-free tasks. Our RND-style bonus, once

its normaliser was guarded, found the reward in 39 of 40 runs but was slower than the penalty alone on every task. The unguarded normaliser is a concrete failure case for small reimplementations: a division by a zero running standard deviation produced an early bonus spike in 20 of 40 runs, and all 19 runs that never found the reward were among them. The noisy TV drew slightly more visits as its token set grew but delayed neither RND nor the observation count, as expected for these two bonuses. We read this as evidence about small deterministic gridworlds and one untuned RND-style implementation; the main follow-ups are stochastic dynamics, a forward-model bonus for the noisy-TV test, and more seeds.

References

- [1] I. Osband, C. Blundell, A. Pritzel, and B. V. Roy, “Deep Exploration via Bootstrapped DQN,” arXiv preprint arXiv:1602.04621, 2016. [Online]. Available: <https://arxiv.org/abs/1602.04621>
- [2] A. L. Strehl and M. L. Littman, “An analysis of model-based Interval Estimation for Markov Decision Processes,” Journal of Computer and System Sciences, 2008. [Online]. Available: <https://doi.org/10.1016/j.jcss.2007.08.009>
- [3] M. G. Bellemare, S. Srinivasan, G. Ostrovski, T. Schaul, D. Saxton, and R. Munos, “Unifying Count-Based Exploration and Intrinsic Motivation,” arXiv preprint arXiv:1606.01868, 2016. [Online]. Available: <https://arxiv.org/abs/1606.01868>
- [4] H. Tang, R. Houthoofd, D. Foote, A. Stooke, X. Chen, Y. Duan, J. Schulman, F. D. Turck, and P. Abbeel, “#Exploration: A Study of Count-Based Exploration for Deep Reinforcement Learning,” arXiv preprint arXiv:1611.04717, 2016. [Online]. Available: <https://arxiv.org/abs/1611.04717>
- [5] G. Ostrovski, M. G. Bellemare, A. van den Oord, and R. Munos, “Count-Based Exploration with Neural Density Models,” arXiv preprint arXiv:1703.01310, 2017. [Online]. Available: <https://arxiv.org/abs/1703.01310>
- [6] D. Pathak, P. Agrawal, A. A. Efros, and T. Darrell, “Curiosity-driven Exploration by Self-supervised Prediction,” arXiv preprint arXiv:1705.05363, 2017. [Online]. Available: <https://arxiv.org/abs/1705.05363>
- [7] Y. Burda, H. Edwards, D. Pathak, A. Storkey, T. Darrell, and A. A. Efros, “Large-Scale Study of Curiosity-Driven Learning,” arXiv preprint arXiv:1808.04355, 2018. [Online]. Available: <https://arxiv.org/abs/1808.04355>
- [8] Y. Burda, H. Edwards, A. Storkey, and O. Klimov, “Exploration by Random Network Distillation,” arXiv preprint arXiv:1810.12894, 2018. [Online]. Available: <https://arxiv.org/abs/1810.12894>
- [9] A. N. Mavor-Parker, K. A. Young, C. Barry, and L. D. Griffin, “How to Stay Curious while Avoiding Noisy TVs using Aleatoric Uncertainty Estimation,” arXiv preprint arXiv:2102.04399, 2021. [Online]. Available: <https://arxiv.org/abs/2102.04399>
- [10] A. A. Taïga, W. Fedus, M. C. Machado, A. Courville, and M. G. Bellemare, “On Bonus-Based Exploration Methods in the Arcade Learning Environment,” arXiv preprint arXiv:2109.11052, 2021. [Online]. Available: <https://arxiv.org/abs/2109.11052>
- [11] J. Martin, S. N. Sasikumar, T. Everitt, and M. Hutter, “Count-Based Exploration in Feature Space for Reinforcement Learning,” arXiv preprint arXiv:1706.08090, 2017. [Online]. Available: <https://arxiv.org/abs/1706.08090>
- [12] T. Rashid, B. Peng, W. Böhmer, and S. Whiteson, “Optimistic Exploration even with a Pessimistic Initialisation,” arXiv preprint arXiv:2002.12174, 2020. [Online]. Available: <https://arxiv.org/abs/2002.12174>
- [13] D. Jarrett, C. Tallec, F. Althé, T. Mesnard, R. Munos, and M. Valko, “Curiosity in Hindsight: Intrinsic Exploration in Stochastic Environments,” arXiv preprint arXiv:2211.10515, 2022. [Online]. Available: <https://arxiv.org/abs/2211.10515>